MODEL ARCHITECHURES FOR COMPARATIVE ANALYSIS:

1. CNN + GRU

As for temporal modelling we used CNN + LSTM architecture, for comparative analysis CNN + GRU could be implemented.

Basically, we would compare LSTM vs GRU under identical feature extarctors and data splits.

Reference papers where GRU is implemented
- https://pubmed.ncbi.nlm.nih.gov/28286600/

Code base:

- PyTorch GRU tutorial:
https://docs.pytorch.org/docs/stable/generated/torch.nn.GRU.html

- https://github.com/rasbt/deeplearning-models

2. TEMPORAL TRANSFORMER

Papers:

- https://www.researchgate.net/publication/365614966_Vision_Transformers_in_Medical_Imaging_A_Review

- https://www.nature.com/articles/s41746-024-01207-4

- https://www.nature.com/articles/s41598-023-30853-z

Code base:

- https://docs.pytorch.org/docs/stable/generated/torch.nn.TransformerEncoder.html

- https://github.com/lucidrains/vit-pytorch

3. 3D CNN / Temporal Convolution (Non-Recurrent)

Instead of treating time as a sequence, we try to treat it as a **third dimension**. For the 3D CNN, we would simply stack 2D B-scans along a new dimension.

Code base:

- https://github.com/kenshohara/3D-ResNets-PyTorch

- https://github.com/locuslab/TCN

4. Similarly, I have talked about two works in my thesis report, it is also possible to adapt their architectures for our 2D B-scans.

- M. Bryant, “Deepmind’s ai detects over 50 eye diseases with 94

- D. Romo-Bucheli, U. Schmidt-Erfurth, and H. Bogunović, “End-to-end deep learning model for predicting treatment requirements in neovascular amd from longitudinal retinal oct imaging,” IEEE Journal of Biomedical and Health Informatics, vol. 24, no. 12, pp. 3399–3408, 2020.

These papers are using the combination of same architectures; I talked in 1-3.

Another paper recommended by Prof. Daniele Meli is

- https://www.nature.com/articles/s41597-024-03844-6.pdf

This paper introduces a dataset for segmentation of wet AMD lesions in OCT which is different from our current task of visual acuity prediction. This can be implemented if your dataset includes annotations for pathology segmentation (fluid, PED, SRF, etc.)